Case Study 29: Password Policy Validation Engine

1. Problem Statement

Design and implement a robust verification engine that validates user-submitted passwords against multi-factor security policies, ensuring compliance with organizational password complexity guidelines.

2. Objectives

- Evaluate passwords against defined security criteria (length, character sets, special symbols).
- Provide detailed feedback for each rule evaluation.
- Determine overall password strength and compliance status.

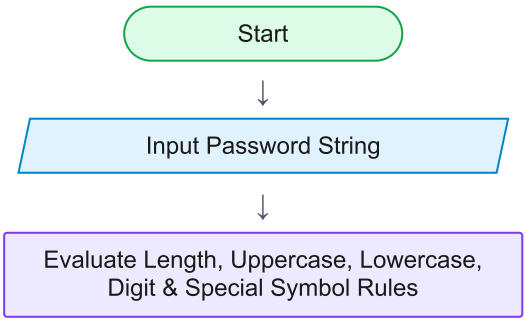
3. Assumptions

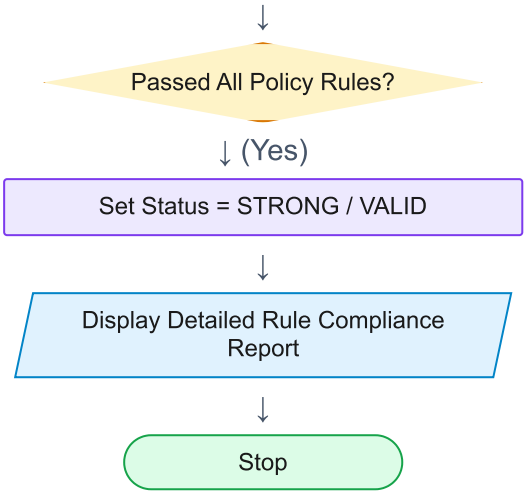
- Minimum password length requirement is **8 characters**.
- Must contain at least one uppercase letter (A-Z), one lowercase letter (a-z), one digit (0-9), and one special character (e.g., `@, $, !, %, *, ?, &, #`).
- Validation returns compliance status for each individual rule as well as an aggregate status.

4. Algorithm

1. Start.
2. Read password string input from user.
3. Check Rule 1: Length >= 8.
4. Check Rule 2: Contains at least one uppercase letter.
5. Check Rule 3: Contains at least one lowercase letter.
6. Check Rule 4: Contains at least one numeric digit.
7. Check Rule 5: Contains at least one special character from `[@$!%*?&#]`.
8. If all rules are satisfied (True), mark password as **STRONG / VALID**.
9. Else, mark password as **WEAK / INVALID**.
10. Display itemized policy evaluation report.
11. Stop.

5. Flowchart





6. Pseudocode

```
START INPUT password rule_len = LENGTH(password) >= 8 rule_upper =
CONTAINS_UPPERCASE(password) rule_lower = CONTAINS_LOWERCASE(password) rule_digit =
CONTAINS_DIGIT(password) rule_special = CONTAINS_SPECIAL(password) is_valid =
rule_len AND rule_upper AND rule_lower AND rule_digit AND rule_special PRINT "Rule
1 (Min Length 8):", rule_len PRINT "Rule 2 (Uppercase):", rule_upper PRINT "Rule 3
(Lowercase):", rule_lower PRINT "Rule 4 (Digit):", rule_digit PRINT "Rule 5
(Special Char):", rule_special IF is_valid THEN PRINT "Final Status: STRONG /
VALID" ELSE PRINT "Final Status: WEAK / INVALID" ENDIF STOP
```

7. Test Case and Expected Output

Test Case 1: Input "Secur3#2026"

```
Validation Results: [PASS] Minimum length of 8 characters [PASS] Contains at least
one uppercase letter [PASS] Contains at least one lowercase letter [PASS] Contains
at least one digit [PASS] Contains at least one special character Final Status:
STRONG / VALID
```

Test Case 2: Input "password123"

```
Validation Results: [PASS] Minimum length of 8 characters [FAIL] Contains at least
one uppercase letter [PASS] Contains at least one lowercase letter [PASS] Contains
at least one digit [FAIL] Contains at least one special character Final Status:
WEAK / INVALID
```

8. Conclusion

The Password Policy Validation Engine enforces authentication security policies by analyzing password strings against key structural criteria, preventing weak password vulnerabilities.